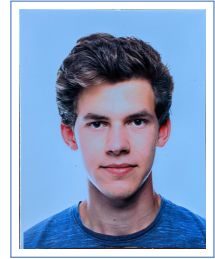


Georg Trede

Physics PhD Student

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Research interests

Scientific machine learning, computational physics and dynamical systems reconstruction, with a focus on non-autonomous systems and generalisation beyond observed dynamical regimes.

Education

- 2025–present **PhD in Physics**, *Heidelberg University*
Supervisor: Prof. Dr. Daniel Durstewitz; Department of Theoretical Neuroscience, Central Institute of Mental Health, Mannheim.
- 2022–2024 **MSc in Physics**, *Heidelberg University*, Final grade: 1.2
Specialisation: Computational Physics. Thesis: *Neural Flow Operators for Solving Advection-Diffusion Systems*.
- 2018–2022 **BSc in Physics**, *University of Münster*, Final grade: 1.5
Interdisciplinary studies: Computer Science. Thesis: *Auto Alignment Toolkit for Optical Resonators*, in collaboration with the Australian National University, Canberra.
- 2021–2022 **Study abroad**, *Australian National University*, Canberra
DAAD scholarship.

Publication

- 2026 **Georg Trede**, Charlotte Ricarda Doll, Elias Weber and Daniel Durstewitz. *Topological Out-of-Domain Generalization in Dynamical Systems Reconstruction*. Preprint, arXiv:2606.22969. Equal contribution with Charlotte Ricarda Doll.

Research experience

- 2023 **Research Assistant**, *Heidelberg University*
Research on neural autoencoders, led by Daniel Durstewitz.
- 2019–2021 **Research Assistant**, *London School of Economics*
Supported Friedrich Geiecke's research project "Populism in Social Media".

Volunteering

Scouting Member of the German Scout Association (DPSG) since 2008; group leader since 2018; head of the local group since 2024. pfadfinder-hd.de

Award

Jugend forscht Special award “Environment” in the state-level competition for “Intelligent Streetlights”, developed with Erik Holzhäuser.

Technical skills and languages

Advanced Python, PyTorch, Linux.

Intermediate HTML, JavaScript, Julia, $\text{\LaTeX}$.

Languages German (native), English (proficient), French (basic).

Additional experience and interests

Other work Part-time work at weekly markets in Münster (2017–2021) and Heidelberg (2022–2024).

Interests Unicycle hockey, time outdoors, reading, building websites, guitar and saxophone.